

Fine-tuning a Small LLM with LoRA for Automated Paper Review Insights

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1 Introduction

Peer review remains a foundational pillar of the scientific process, ensuring rigor, validity, and quality through independent expert evaluation (Bornmann, 2011). However, as the number of submissions to journals and conferences continues to grow, the peer review system faces increasing strain, leading to delayed editorial decisions, inconsistent feedback, and reviewer fatigue (Aczel et al., 2025). In parallel, recent advancements in Large Language Models (LLMs) have opened new opportunities for assisting or partially automating aspects of the review process. Current approaches vary in cost, flexibility, and performance, and include: (1) prompt-based zero-shot generation, (2) full supervised fine-tuning on human review data, and (3) parameter-efficient fine-tuning such as Low-Rank Adaptation (LoRA). Prompt engineering involves using general-purpose LLMs (e.g., GPT-4) in a zero-shot setting to generate review-like text given paper abstracts and task-specific instructions (Gilardi et al., 2023). This approach is simple and requires no additional training, but its outputs are highly sensitive to prompt design and lack domain adaptation. In contrast, full supervised fine-tuning retrains all parameters of an LLM on large corpora of human-written reviews, producing highly specialized models (Zhou et al., 2022). While this method can achieve strong task-specific performance, it demands significant computational resources and large-scale annotated datasets. Parameter-efficient fine-tuning techniques, such as LoRA (Hu et al., 2021), offer a compromise by updating only a small number of low-rank parameters within the model, enabling task adaptation even with limited data and compute.

This project investigates whether LoRA fine-tuning can improve the ability of a small open-source LLM—specifically LLaMA 3.2B Instruct model—to extract structured strengths and weaknesses from scientific manuscripts derived from the OpenReview platform, and whether such fine-tuning influences the model’s ability to estimate numeric review scores. The overarching aim is to build a lightweight, fully reproducible system capable of producing interpretable, structured reviewer-style output drawn from abstracts, titles, and full review text. Our evaluation compares the zero-shot behaviour of the base model against its LoRA-tuned counterpart, using both regression metrics for numeric ratings and semantic measures for the quality of extracted strengths and weaknesses. By examining these two paradigms under controlled conditions, we aim to evaluate whether parameter-efficient fine-tuning constitutes an advantage in low-resource environments or whether prompt conditioning alone is sufficient.

2 Methods

The dataset used in this study was assembled from the OpenReview platform, where both peer reviews and associated metadata are publicly accessible. Text fields—including titles, abstracts, and review

⁰Project repository on GitHub: https://github.com/camillabonomo02/Automated_Paper_Review.git

bodies—were cleaned by removing URLs, normalizing whitespace, and stripping templated markup. Ratings were extracted from heterogeneous formats and converted to a consistent 1–10 numerical scale. The cleaned dataset was divided into training and validation partitions, with care taken to preserve the distribution of scores.

A crucial component of the methodology involves the creation of structured training targets for strengths and weaknesses, because human-written reviews rarely present these elements as neatly formatted bullet lists. To address this, a zero-shot “teacher” model was prompted to generate canonical JSON objects containing strengths and weaknesses exclusively from the title and abstract of each paper. This approach yielded a synthetic but consistent supervision signal. Since automatically generated data can be noisy or malformed, the system incorporated robust parsing, retry attempts under stricter prompting, and fallback mechanisms guaranteeing the presence of at least one non-empty strength and weakness. To reduce computational burden, only a subset of the validation set was processed by the teacher model, while the full training set was used after the introduction of resume-based continuation for long-running generation jobs.

LoRA fine-tuning was then applied to the base LLaMA 3.2B model. Only a small portion of the model’s projection matrices was updated, allowing specialised learning without modifying the bulk of the model’s parameters. The fine-tuning objective was intentionally narrow: the model was trained to reproduce the teacher-generated structured JSON, rather than to generate full reviews or infer numeric ratings. This deliberate constraint was intended to isolate the effect of LoRA on the structure and content of strengths and weaknesses, without blending it with other tasks.

In parallel, rating prediction was handled separately. Instead of training a new regression head, the model was prompted to output only a numeric rating, and the log-probabilities of candidate digits 1 through 10 were extracted from the model’s token distribution. These probabilities were used to compute an expected value, which served as the predicted rating. Because uncalibrated logits often lack a meaningful scale, a Huber regression model was fitted to map predicted values onto the distribution of human-annotated ratings, using a subsampled portion of the training data to avoid excessive computational costs.

Evaluation proceeded along two complementary dimensions. For rating prediction, traditional regression metrics—MAE, RMSE, and Pearson correlation—were computed before and after calibration to determine whether the model’s numeric judgements aligned with human reviewers. For strengths and weaknesses, we measured fallback frequency, assessed lexical diversity across all generated bullets, and computed the cosine similarity between embeddings of teacher-generated and model-generated outputs. A small qualitative inspection was performed to understand failure patterns and identify cases where fine-tuning improved or degraded the informativeness of the extracted content.

3 Results

The two models—zero-shot and fine-tuned—exhibited markedly different behaviour on the numeric rating task (Table 1). Before calibration, the zero-shot model produced moderately correlated predictions with reasonably controlled error, whereas the fine-tuned model displayed significantly larger errors and an almost negligible correlation with human scores. After calibration, both models improved substantially, but the calibrated zero-shot model remained distinctly superior: its MAE fell to approximately 1.8, and its RMSE to just above 2.0, while preserving a Pearson correlation of around 0.36. By contrast, the calibrated fine-tuned model achieved higher MAE and RMSE and showed no positive correlation with the ground truth. This divergence suggests that fine-tuning exclusively on strengths and weaknesses subtly disrupted the model’s internal representations used for judging overall quality.

On the structured extraction task, the contrast between the two models was much smaller (Table 2). Both zero-shot and fine-tuned models produced valid JSON for every example, with no observed fallback. However, the zero-shot model achieved higher lexical diversity and stronger semantic similarity to the teacher outputs. Manual inspection reinforces this observation. For some papers, such as those discussing higher-order function networks, the fine-tuned and zero-shot outputs were nearly indistinguishable,

Task	MAE	RMSE	Pearson
Zero-shot raw	2.72	3.26	0.36
Fine-tuned raw	3.05	3.69	0.07
Zero-shot calibrated	1.82	2.06	0.36
Fine-tuned calibrated	1.99	2.19	-0.07

Table 1: *Regression metrics before and after calibration*

suggesting that the base model was already capable of performing the task without explicit adaptation. In more challenging cases, fine-tuning sometimes led to shorter, less detailed summaries that resembled fallback-like behaviour, even though no fallback mechanism was triggered. This result again indicates that the fine-tuned model may have over-fit to the synthetic training distribution, producing overly homogenized output despite being trained on structurally valid examples.

Task	Fallback rate	Lexical div.	Teacher sim. cos.
Zero-shot	0.0	0.23	0.54
Fine-tuned	0.0	0.19	0.51

Table 2: *Semantic evaluation of models*

Overall, the results point toward an asymmetric conclusion: while zero-shot prompting already provides surprisingly strong structured extraction and moderately reliable rating estimation, the LoRA fine-tuned model did not offer meaningful improvements and in fact performed worse on the rating task.

4 Final Discussion

The findings of this study suggest that parameter-efficient fine-tuning, at least in the low-resource configuration examined here, does not reliably improve structured peer-review extraction relative to a carefully designed zero-shot prompting strategy. The base LLaMA model already possesses substantial prior knowledge about academic writing conventions and appears capable of identifying key strengths and weaknesses directly from abstracts. As a result, fine-tuning on a relatively small, synthetic dataset of strengths and weaknesses did not meaningfully expand the model’s capabilities and occasionally produced shallower or more repetitive summaries.

Several strategies were explored to mitigate these limitations, including the idea of increasing the amount of synthetic data used for fine-tuning (for instance, extending the number of teacher-labeled examples from 500 to 1,000) and adjusting the number of fine-tuning epochs. Intuitively, exposing the model to more examples should improve robustness and reduce overfitting to idiosyncratic patterns. However, a critical constraint of the current setup is that the teacher remains the same underlying model. If the teacher’s strengths and weaknesses are only moderately informative—correct in a broad sense but not particularly insightful—then the fine-tuned model can, at best, learn to imitate that level of quality. Increasing the number of training examples primarily increases coverage of the teacher’s behaviour; it does not fundamentally improve the quality of the supervision signal. In other words, LoRA fine-tuning in this setting is limited by a “ceiling effect” imposed by the teacher’s own performance.

This points to a deeper methodological limitation: the model is being fine-tuned to predict strengths and weaknesses without access to any genuine human ground truth for S/W. The supervision signal is entirely synthetic, derived from another LLM that is subject to its own biases, omissions, and occasional hallucinations. From a statistical perspective, this creates a form of “teacher–student loop” in which the student model is optimized to approximate the teacher’s outputs rather than to approximate human judgments directly. Any systematic errors or stylistic quirks of the teacher become baked into the fine-tuned model. As a consequence, the evaluation of the fine-tuned model on S/W extraction is, to some extent, circular: improved similarity to the teacher does not necessarily imply better alignment with human reviewers, and may even move the model away from the nuances present in real peer-review texts.

The degradation observed in rating prediction after fine-tuning further illustrates how targeted adaptation can have unintended side effects. Even though the fine-tuning objective did not involve ratings, updating internal representations to better mimic the teacher’s S/W patterns appears to have disrupted latent dimensions used for quality assessment. This suggests that multi-task learning, or at least explicitly including rating supervision in the fine-tuning objective, may be necessary if future work aims to unify structured extraction and scoring within a single adapted model. Otherwise, there is a real risk that improving performance on one synthetic objective (matching teacher-style S/W) comes at the expense of performance on a more meaningful downstream metric (correlation with human scores).

Several limitations therefore temper the generality of the conclusions. The synthetic teacher labels may fail to capture the full diversity of human reviewing practices, and the relatively small dataset limits both the richness of the supervision signal and the stability of the fine-tuning process. The model used is also relatively small, and larger models might either benefit more from LoRA or exhibit different trade-offs between structure, content, and rating prediction. Moreover, the absence of human-annotated strengths and weaknesses means that, while the pipeline is fully reproducible and computationally accessible, it cannot yet claim to optimize for human-aligned evaluation criteria.

Despite these constraints, the study provides useful insights for the design of future systems. One natural direction is to integrate even a small amount of human-labeled S/W as an anchor, using LLM-generated labels only as a supplement rather than as the sole source of supervision. Another is to adopt a genuinely multi-task setup, where the model is trained simultaneously on structured extraction and numeric scoring, reducing the risk that improvements in one subspace damage performance in another. Finally, more sophisticated calibration methods and constrained decoding strategies could help stabilize both rating prediction and JSON generation, especially when models are deployed in real editorial workflows.

In conclusion, the present experiments show that zero-shot prompting combined with logit-based rating prediction and lightweight calibration already constitutes a strong baseline for automated peer-review assistance, whereas LoRA fine-tuning on synthetic strengths and weaknesses provides limited added value under the current conditions. The work highlights both the promise and the limits of teacher–student pipelines without human ground truth, and underscores the importance of aligning future fine-tuning efforts with evaluation criteria that reflect actual human reviewing practice.

5 References

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